

Gastrointestinal symptoms in 3181 volunteers ingesting snack foods containing olestra or triglycerides. A 6-week randomized, placebo-controlled trial.

BACKGROUND: Olestra is a nonabsorbable, energy-free fat substitute. Because it is not absorbed, it may cause digestive symptoms when consumed in large amounts. **OBJECTIVE:** To compare the frequency and impact of gastrointestinal symptoms in adults and children who freely consume snacks containing olestra or regular snacks in the home. **DESIGN:** 6-week, double-blind, randomized, parallel, placebo-controlled trial. **SETTING:** General community. **PARTICIPANTS:** 3181 volunteers 2 to 89 years of age. **INTERVENTION:** Households received identical packages labeled as containing olestra corn or potato chips. These packages contained either olestra or regular chips (control). **MEASUREMENT:** Gastrointestinal symptoms and their impact on daily activities were reported in a daily record. **RESULTS:** At least one gastrointestinal symptom was reported by 619 of 1620 (38.2%) persons in the olestra group and 576 of 1561 (36.9%) controls (difference, 1.3 percentage points [95% CI, -3.6 to 6.2 percentage points]; $P = 0.60$). In general, the groups did not differ significantly in the proportion of participants who reported individual gastrointestinal symptoms; however, more controls reported nausea (8.4% compared with 5.7%; difference, -2.7 percentage points [CI, -4.9 to -0.4 percentage points]; $P = 0.02$). The only difference between groups for the mean numbers of days on which symptoms were reported was that participants in the olestra group had 1 more symptom-day of more frequent bowel movements than did controls (3.7 symptom-days compared with 2.8 symptom days; difference, 0.9 symptom-days [CI, 0.1 to 1.8 symptom-days]; $P = 0.04$). The groups did not differ in the impact of symptoms on daily activities. **CONCLUSIONS:** Clinically meaningful or bothersome gastrointestinal effects are not associated with unregulated consumption of olestra corn and potato chips in the home.